Do not write outside the box

0 4 This question is about Group 7 elements.

0 4 . 1 What are the Group 7 elements known as?

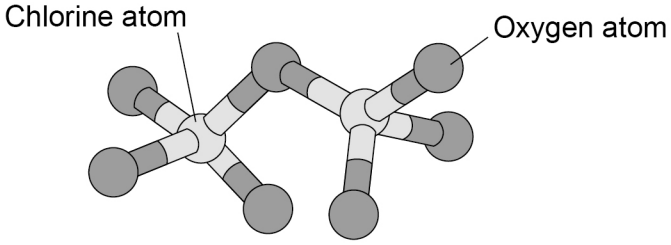
[1 mark]

0 4 . 2 Why do Group 7 elements react in similar ways?

[1 mark]

0 4 . 3 **Figure 3** shows the structure of a molecule of chlorine oxide.

Figure 3



What is the molecular formula of the chlorine oxide molecule in **Figure 3**?

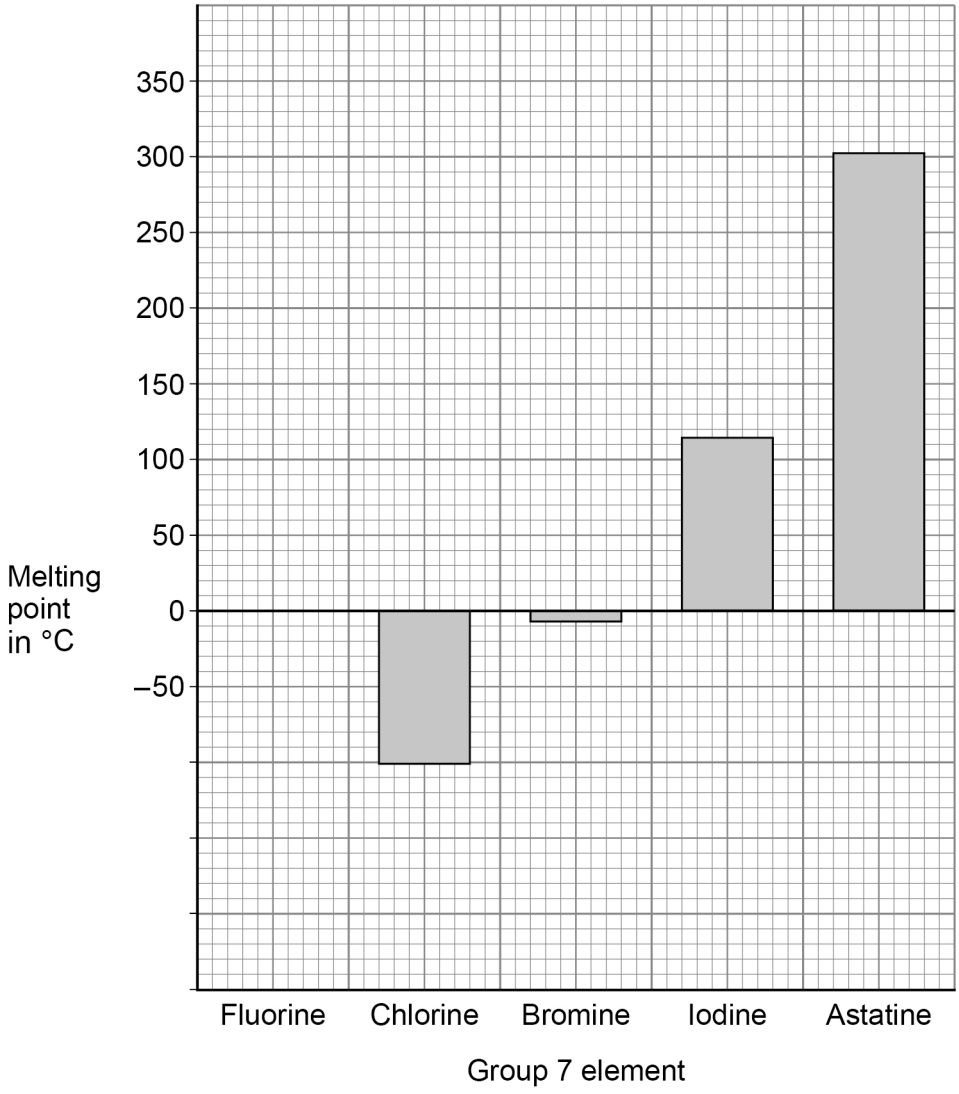
[1 mark]

Question 4 continues on the next page

Do not write outside the box

Figure 4 shows the melting points of some Group 7 elements.

Figure 4



0 4 . 4 The melting point of fluorine is –220 °C

Complete **Figure 4**.

You should:

- complete the scale on the y-axis
- draw the bar for the melting point of fluorine.

[2 marks]

Do not write outside the box

0 4 . 5 Explain the trend in the melting points of the Group 7 elements.

Use **Figure 4**.

[3 marks]

0 4 . 6 What is the state symbol for bromine at –50 °C?

Use **Figure 4**.

[1 mark]

Tick (✓) **one** box.

(aq) ☐ (g) ☐ (l) ☐ (s) ☐

0 4 . 7 Evaporation and boiling occur at the surface of bromine at its boiling point.

Name **one** more process that happens at the surface of bromine at its boiling point.

[1 mark]

10

Turn over for the next question